

Class 10 Physics

ICSE and CBSE Board Syllabus

ICSE Syllabus

Theme 1: Force, Work, Power and Energy

- Moment of force, equilibrium, centre of gravity.
- Uniform circular motion.
- Work, energy and power relations; different forms of energy.

Theme 2: Machines

- Load, effort, mechanical advantage, velocity ratio and efficiency.
- Levers and pulley systems with simple numericals.

Theme 3: Refraction of Light

- Refraction through glass slab / prism basics, refractive index, real and apparent depth.

Theme 4: Refraction through a Lens

- Convex and concave lenses, image formation, lens formula and magnification.

Theme 5: Spectrum

- Dispersion, pure spectrum, electromagnetic radiation basics, scattering of light / colour phenomena.

Theme 6: Sound

- Wave nature, amplitude, frequency, wavelength, pitch, loudness, echo and resonance.

Theme 7: Current Electricity

- Charge, current, potential difference, resistance, Ohm's law, series/parallel combination, heating effect.

Theme 8: Household Electricity & Power

- Electrical power and energy, commercial unit, domestic circuits, fuse / MCB / earthing.

Theme 9: Magnetism and Electromagnetism

- Magnetic field, electromagnets, electric motor / generator basics, electromagnetic induction.

CBSE Syllabus

Unit 1: Light – Reflection and Refraction

- Reflection by spherical mirrors, mirror formula, magnification.
- Refraction through glass slab and lenses, lens formula, power of lens.

Unit 2: Human Eye and the Colourful World

- Defects of vision and correction.
- Dispersion, atmospheric refraction, scattering.

Unit 3: Electricity

- Current, potential difference, resistance.
- Ohm's law, resistors in series and parallel.
- Heating effect of current, electric power and energy.

Unit 4: Magnetic Effects of Electric Current

- Magnetic field due to current, right-hand thumb rule.
- Force on current-carrying conductor, electric motor.
- Electromagnetic induction and domestic electric circuits.

Unit 5: Sources of Energy

- Conventional and non-conventional sources, environmental impact, efficiency and sustainability.